

Evaluating cessation of the type 2 oral polio vaccine by modeling pre- and post-cessation detection rates



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ABSTRACT

The globally synchronized removal of the attenuated Sabin type 2 strain from the oral polio vaccine (OPV) in April 2016 marked a major change in polio vaccination policy. This change will provide a significant reduction in the burden of vaccine-associated paralytic polio (VAPP), but may increase the risk of circulating vaccine-derived poliovirus (cVDPV2) outbreaks during the transition period. This risk can be monitored by tracking the disappearance of Sabin-like type 2 (SL2) using data from the polio surveillance system. We studied SL2 prevalence in 17 countries in Africa and Asia, from 2010 to 2016 using acute flaccid paralysis surveillance data. We modeled the peak and decay of SL2 prevalence following mass vaccination events using a beta-binomial model for the detection rate, and a Ricker function for the temporal dependence. We found type 2 circulated the longest of all serotypes after a vaccination campaign, but that SL2 prevalence returned to baseline levels in approximately 50 days. Post-cessation model predictions identified 19 anomalous SL2 detections outside of model predictions in Afghanistan, India, Nigeria, Pakistan, and western Africa. Our models established benchmarks for the duration of SL2 detection after OPV2 cessation. As predicted, SL2 detection rates have plummeted, except in Nigeria where OPV2 use continued for some time in response to recent cVDPV2 detections. However, the anomalous SL2 detections suggest specific areas that merit enhanced monitoring for signs of cVDPV2 outbreaks.

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1. Introduction

Since the introduction of the oral polio vaccine (OPV), OPV use has eliminated the three serotypes of wild poliovirus from all but three countries. Of the three serotypes, only wild type 1 is known to be circulating in those countries; wild type 2 has been eradicated globally and wild type 3 has not been detected since 2012.

The effectiveness of OPV comes from its abilities to prevent poliomyelitis, induce intestinal mucosal immunity which prevents reinfection, and transmit to and immunize contacts of vaccine recipients [1]. However, the live-attenuated Sabin strains in OPV

(Sabin 1, 2, and 3) are genetically unstable, and evolve in humans first into Sabin-like (SL) viruses and then, on rare occasions, into circulating vaccine-derived polioviruses (cVDPV) that are phenotypically indistinguishable from wild-type poliovirus. The loss of genetic attenuation in Sabin viruses can cause vaccine-associated paralytic poliomyelitis (VAPP) in vaccine recipients and their close contacts or outbreaks of cVDPVs in under-immunized communities [2]. In particular, the type 2 component of OPV has caused ~90% of all VDPV cases (~45 cases annually) and 26–31% of all VAPP cases (~100 to 200 cases annually) [3]. OPV or SL exposure can also lead to prolonged infection and shedding in immunodeficient individuals, in some instances leading to immunodeficient VDPV (iVDPV) [3].

As wild type 2 poliovirus has been eradicated, the type 2 component of OPV was withdrawn from use globally in April 2016 (OPV2 cessation) to eliminate the burden of type 2 VAPP and cVDPV2. This change will significantly reduce the burden of VAPP, but SL2 that was circulating prior to OPV2 cessation could continue to circulate and lead to outbreaks of cVDPV2. When such outbreaks occur, OPV2 response campaigns will be necessary to stop them.

Abbreviations: cVDPV, circulating (demonstrating sustained person-to-person transmission) vaccine-derived poliovirus; mOPV2, monovalent oral polio vaccine containing only Sabin 2; NP-AFP, non-polio acute flaccid paralysis; OPV, oral polio vaccine; RI, routine immunization; SIA, supplemental immunization activity; SL2, Sabin-like type 2 viruses; tOPV, trivalent oral polio vaccine containing Sabin 1, 2, and 3; VAPP, vaccine-associated paralytic polio; VDPV, vaccine-derived polioviruses.

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Unfortunately, the use of OPV2 will lead to additional SL2 transmission which could develop into secondary cVDPV2 outbreaks outside of the response region. This paper aims to assess the progress of OPV2 cessation by setting benchmarks for the duration and prevalence of SL2 detection post-cessation based on pre-cessation SL2 isolation rates in the stool of children monitored by the polio surveillance system.

Prior to OPV2-cessation, isolation of SL2 in stool could be attributed to vaccination through routine immunization (RI), mass supplemental immunization activities (SIAs), or their associated secondary transmission. RI reaches children before 12 months of age with multiple doses of trivalent OPV (tOPV, containing all three Sabin vaccines), and this creates a baseline prevalence that is roughly constant over time. In contrast, SIAs reach large numbers of 0–5 year-old children during short, 3–5 day campaigns that cause pulses or spikes in SL detection rates. Within countries at risk of polio outbreaks or importations, SIAs occur multiple times per year and their geographic extents can range from a single district to an entire country [4]. Various formulations of OPV have been used in SIAs: tOPV, bivalent OPV (bOPV containing Sabin 1 and 3), or monovalent OPV (mOPV, containing any one of the Sabin types). After April 2016, mOPV2 is only intended to be used in response to cVDPV2 outbreaks, with bOPV substituted for tOPV in RI with the addition of inactivated polio vaccine (IPV).

The duration of SL2 detection after cessation will depend on viral shedding from primary OPV recipients, their contacts, the contacts of their contacts, etc. Shedding in primary recipients mostly occurs within 3–4 weeks of vaccination, but can sometimes persist for 7 weeks or more [5,6], and shedding duration tends to decrease with increasing immunity from prior poliovirus exposure [1,5]. At the community level, after local OPV2 cessation, settings with sensitive environmental surveillance have shown that SL can remain detectable for between 5 weeks [7,8] and 6 months [9–13]. The difference in duration of detection between primary recipients and community surveillance provides a measure of the role of transmission in SL persistence [14,15].

To evaluate the prevalence and duration of SL2 detection after OPV2 cessation, we compared SL1, 2, and 3 detection rates in the years prior to and immediately after OPV2 cessation in 17 countries in western Africa and southern Asia. Specifically, for each type we assessed (1) the duration of the pulse in SL detection rates after SIAs; (2) the baseline detection rates between SIAs; and (3) detection rates after type 2 was removed from circulation. In combination, these objectives allow us to develop benchmarks for the successful cessation of OPV2 vaccination and speculate on the risk for future cVDPV2s following anomalous SL2 detection. In addition, our analyses provide expectations for SL2 detection after OPV2 use in outbreak response.

2. Materials and methods

2.1. Data

SL detection rates were based on acute flaccid paralysis (AFP) surveillance case data maintained globally by the World Health Organization (WHO) [16]. When an AFP case is reported, along with clinical and background information, two stool samples are collected within 14 days of each other and tested for poliovirus by WHO-accredited laboratories in the Global Polio Laboratory Network. We used AFP cases that are not poliomyelitis (NP-AFP) as the basis of this analysis, as lab testing also flags the presence of SL strains in stool [17]. Typically, the cause of paralysis is not the isolated SL virus. Instead, SL detection acts as a measure of prevalence among the sampled population. NP-AFP cases are often considered to be an age-structured random sample of the popula-

tion [18]. An isolate is classified as SL (Sabin-like) if it is similar to the parent Sabin strain, having not passed a type-specific threshold of genetic divergence indicating classification as a VDPV [19].

Permission for access to the Polio Information System (POLIS) database containing AFP and SIA data was granted to the Institute for Disease Modeling researchers through the World Health Organization. Institutional ethics approval was not sought for AFP surveillance data as they are retrospective and anonymized.

We analyzed NP-AFP data collected from January 2010 to December 2016 in 17 countries that were among the last strongholds of wild poliovirus and have had mass vaccination campaigns: Afghanistan, India, Pakistan, Nigeria, and 13 countries grouped as western Africa (see [section A in the Supplemental Material](#) for inclusions). We parameterized our models using all NP-AFP data between the first reported SL detection within each country ([Table A in the Supplemental Material](#)) and March 1, 2016, just prior to OPV2 cessation which occurred on April 30, 2016. We chose the cutoff of March 2016 to examine model predictions for the final SIAs before cessation. For reference, [Table 1](#) includes a summary of the number of cases, number of OPV positives, dose distribution, and SIA frequency by region.

We obtained supplemental immunization activity (SIA) start dates and location information for SIAs from the WHO and supplemented it with information from country polio eradication programs (Afghanistan, Pakistan, & Nigeria) and published vaccination announcements (India). Our analysis of India was restricted to Bihar and Uttar Pradesh provinces since these areas report the majority of NP-AFP and verifiable campaign data in India. Pattern matching functions were used to infer uncertain campaign start dates in India (see [section B in the Supplemental Material](#)).

2.2. Matching NP-AFP cases and OPV campaign dates

For each NP-AFP case, we calculated the time between stool collection and potential last exposure to each OPV type from SIAs by matching the district location of the case to the SIA calendar. The NP-AFP case might have been exposed to OPV viruses as a primary recipient or through secondary spread. Discussion of possible errors in recorded SIA start dates may be found in [section D in the Supplemental Material](#).

2.3. Modeling the probability of detecting OPV after SIAs

Our approach consists of first modeling patterns of SL detection in NP-AFP as a function of days since the last SIA, and second using predicted SL detection rates from this model to study isolations of SL after OPV2 cessation.

We modeled the relationship between SL detection rates in NP-AFP and days since the last SIA by region and OPV type using a Ricker function. The Ricker function captures a rapid initial rise and peak in SL detection rates after mass vaccination through SIAs that gradually return to baseline levels. We accounted for overdispersion in the outcomes using a Beta-Binomial likelihood to estimate the components of the detection probability p_{ijt} , indexed by region i , SL type j , and t days from the last SIA, that is

$$Y_{ijt} \sim \text{BetaBinomial}(n_{ijt}, p_{ijt}, \theta_{ij}), \text{ with}$$

$$p_{ijt} = \alpha_{ij} + \beta_{ij} t e^{-\gamma_{ij} t}.$$

Here Y_{ijt} represents the number of SL positives of n_{ijt} NP-AFP cases, θ_{ij} the overdispersion, α captures baseline detection rates (likely coming from routine immunization), and β_{ij} and γ_{ij} parameters to describe the shape of the curve. We estimated model parameters with a Bayesian approach, with posterior samples generated

Table 1

Summary of AFP data and vaccination coverage by region. For calculated values, we list the median with the inner quartile range in parentheses. N represents the number of NP-AFP cases. SL positives are the count of NP-AFP cases positive for each Sabin type. Doses represent the parent-reported number of doses. DTP3 represents WHO-UNICEF estimates for three dose diphtheria, tetanus, and pertussis vaccine coverage (co-administered with OPV in infants) in 2015 [32]. OPV/year represents the mean number of SIAs by OPV type that occurred in the year prior to each AFP case (regardless of child age). Campaign duration is in days. The time period of the data used in our models for each region is presented in Table A in the Supplemental Material. OPV = oral poliovirus vaccine. SIA = supplementary immunization activity. SL = Sabin-like poliovirus.

Region	N	SL1 positive	SL2 positive	SL3 positive	Doses	DTP3	OPV1/yr	OPV2/yr	OPV3/yr	Campaign duration
Afghanistan	13207	503	236	548	12 (8–18)	78	5 (4–7)	2 (2–2)	5 (4–7)	3 (3–3)
India	67036	1552	1064	1671	20 (12–29)	87	5 (5–6)	2 (2–2)	5 (5–6)	5 (4–6)
Nigeria	55171	2456	1895	2926	6 (4–10)	56	5 (3–8)	2 (1–3)	5 (3–8)	4 (4–4)
Pakistan	33049	968	514	1062	11 (9–15)	72	9 (7–12)	2 (2–3)	9 (6–11)	3 (3–4)
Western Africa	10689	518	345	457	4 (3–6)	79	4 (2–5)	2 (1–2)	4 (2–5)	4 (3–5)

through Markov Chain Monte Carlo (MCMC) as implemented in STAN v2.10.0 via the rstan package [20] in R v3.3.2 [21]; further details may be found in section E in the Supplemental Material.

The peak detection rate is given by $\alpha_{ij} + (\beta_{ij}/\gamma_{ij})e^{-1}$. We define the time for detection rates to reach baseline levels after SIAs as the time at which predicted detection probabilities (without α_{ij}) declined to 1% of peak detection rates. The timescale of decay in detection rates after peak detection is given by γ_{ij}^{-1} .

We generated 95% credible intervals for each parameter and relevant differences between parameters from posterior samples. A difference was considered significant if the corresponding credible intervals did not contain zero. For plotting and additional model checks we generated prediction intervals for observed pre-cessation rates of SL detection by sampling from a beta-binomial distribution defined by the posterior parameter samples.

The compatibility of predicted and observed post-cessation SL detection rates was evaluated with out-of-sample predictions. Due to OPV2 cessation, we assumed that baseline detection of SL2 (that is, α_{ij}) would decline after April 2016 according to γ_{ij} , i.e. $p_{ijt} = \alpha_{ij}e^{-\gamma_{ij}t} + \beta_{ij}te^{-\gamma_{ij}t}$ where x is the time since cessation. Daily post-cessation 95% prediction intervals were generated by simulating possible SL2 detections for actual post-cessation NP-AFP cases, based on the predicted probability of SL2 detection computed with the SIA exposures of the cases and posterior parameter samples (section G in the Supplemental Material). Several provinces in Nigeria (Adamawa, Bauchi, Borno, Gombe, Jigawa, Sokoto, Taraba, and Yobe) and Niger (Diffa) were excluded from the posterior predictive analysis due to their use of mOPV2 (after OPV2 cessation) in response to a pre-cessation cVDPV2 outbreak in Borno.

3. Results

Inspection of SL detection rates in NP-AFP data (Fig. 1) and its aggregation by time since an OPV SIA (Fig. 2) demonstrated that SL detection rates pulsed rapidly and briefly after SIAs. Detection rates between pulses from SIAs reflect low prevalence and were roughly uniformly distributed in time, compatible with detecting SL from routine immunization.

3.1. Duration of pulse

The pulse in SL2 detection rates after SIAs lasted less than 50 days for all Sabin types (Figs. 2, 3a). The mean duration of the SL2 pulse in detection rates was 13.8 (95% CI 6.68–21.86) days longer than SL1 in all regions except for western Africa (Fig. 1 in the Supplemental Material). The SL2 pulse also tended to be longer than that of SL3, but only significantly so in Nigeria. The pulse in SL3 detection tended to last longer than SL1, but only significantly so in Afghanistan and India.

3.2. Rate of decay

The decay rate (γ) describes the exponential-like rate of decline in SL detection after the post-SIA peak. SL2 detection rates declined more slowly than SL1 within all regions except western Africa (Fig. 3b, Fig. 1 in the Supplemental Material). SL2 detection rates also tended to decline more slowly than SL3, but only significantly so in Nigeria. SL3 tended to decline more slowly than SL1, but only significantly so in Afghanistan and India.

3.3. Peak detection rates

Peak detection rates among the SL types were similar in each region (Fig. 3c), except in India where type 3 peak detection rates were lower than those for type 1 and 2.

3.4. Baseline detection rates

Baseline detection rates (>50 days after SIAs) of SL2 were lower than SL1 & 3 in all regions, except for western Africa (Fig. 3d, Fig. 1 in the Supplemental Material). SL1 and 3 baseline detection rates were not significantly different except in Nigeria where SL3 baseline detection rates were higher than SL1.

3.5. Detection rates after OPV2 cessation

Post-cessation model predictions estimated the probability of detecting SL2 in a NP-AFP case to be less than 0.001 after June 2016, but SL1 and 3 detection rates persisted unchanged (Fig. 4). Observed detection rates largely agreed with these predictions—SL2 detection rates declined dramatically after OPV2 cessation in April 2016, while SL1 & 3 rates were consistent with pre-cessation detection rates. We examined all 19 SL2 detections that occurred outside the 95% prediction intervals after cessation (0.01% of post-cessation observations) since each anomalous SL2 detection may be a potential indicator of prolonged transmission or unsanctioned tOPV use, even though approximately 5% of events will naturally be outside of those intervals. In comparison, we observed 3.3% and 4.1% of SL1 and 3 observations outside of the 95% prediction intervals, respectively, during the same period. Ten of the SL2 detections occurred in India, Nigeria, and Pakistan (three, four, and three, respectively) less than 50 days after cessation – compatible with shedding from primary or secondary contact with OPV2 just prior to cessation. One anomalous case occurred in Kandahar, Afghanistan 141 days after cessation. The detection in western Africa occurred in Guinea 63 days after the last scheduled SIA and 58 days after cessation. Nigeria (excluding the provinces with ongoing mOPV2 use) had seven SL2 detections more than 50 days after cessation. These detections occurred more than 125 days after the last scheduled OPV2 SIAs in their respective districts and 50–450 km from the nearest areas where mOPV2 was used. Thus, these seven detections either resulted from long-term local transmission after cessation, long-distance exportations of SL2 outside of the mOPV2 usage area, or unsanctioned tOPV use.

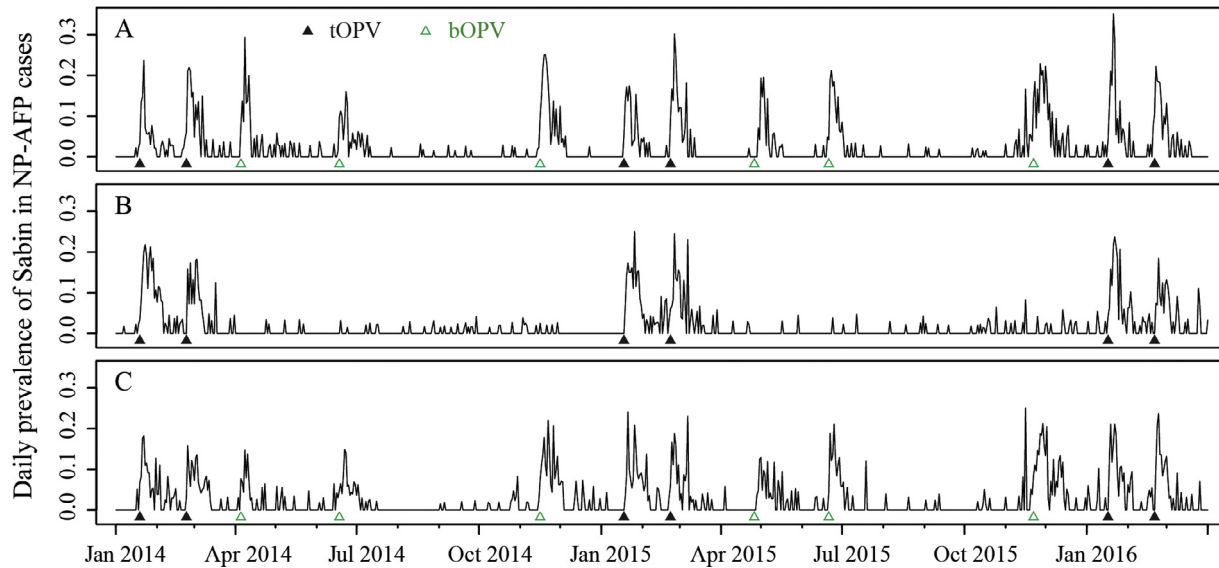


Fig. 1. Daily prevalence of SL1–3 over time in Bihar province, India. The proportion of NP-AFP children positive for SL1 (A), 2 (B), and 3 (C) are shown by day of stool collection (solid lines). Triangles at the bottom of the panels show when SIAs were done, with the colors indicating vaccine used (tOPV containing OPV1, 2, and 3 in black, and bOPV containing OPV1 and 3 in green). OPV detection rates pulse very rapidly and briefly after SIAs. Between campaigns, detection rates are sporadic and of low prevalence, compatible with detecting SL from routine immunization that is continually being administered to children <1 year of age. NP-AFP = non-polio acute flaccid paralysis. OPV = oral poliovirus vaccine. SIA = supplementary immunization activity. SL = Sabin-like poliovirus. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

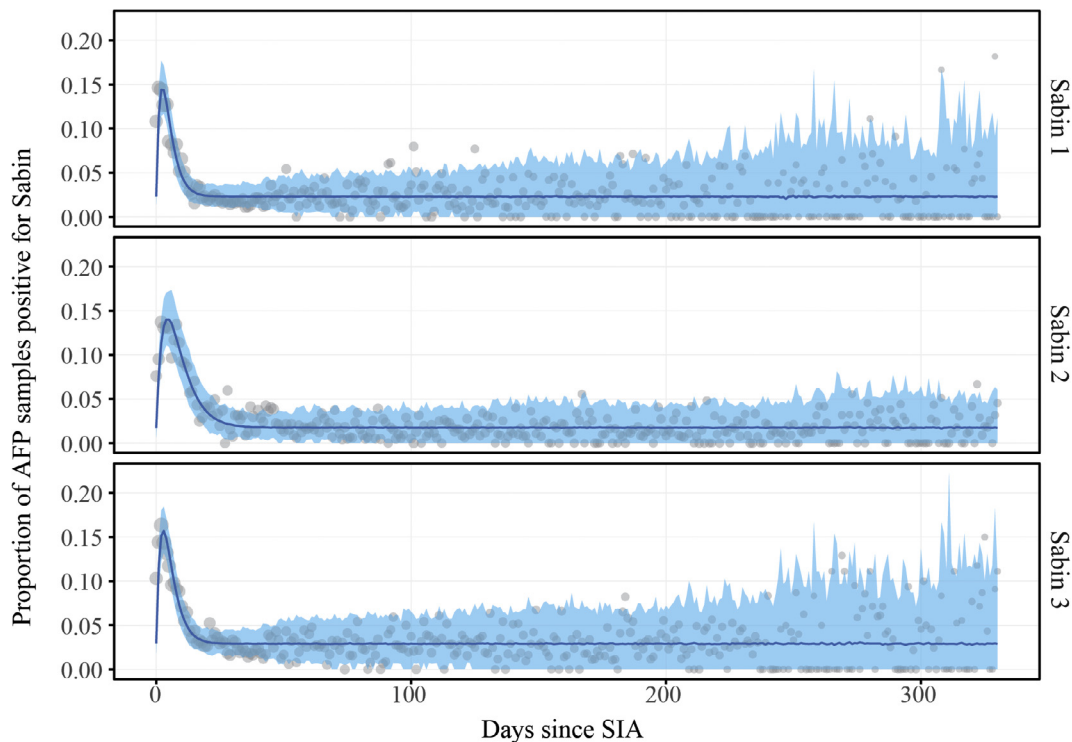


Fig. 2. Pre-cessation Sabin-like 1–3 (SL1–3) detection rates grouped by days since mass supplemental immunization activities (SIA) in Nigeria. The grey dots represent observed rates of Sabin detection and are scaled to the square root of the number of samples. The solid blue line represents mean model predictions and the blue ribbon represents 95% prediction intervals. Figures for the other regions are shown in the Fig. H in the Supplemental Material. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

4. Discussion

4.1. Primary recipients dominate detection of OPV after SIAs

Our models indicated that the brief pulses in pre-cessation detection rates of all SL types after SIAs are compatible with shed-

ding from primary vaccine recipients. We found that the AFP detection rates of all Sabin types declined to baseline rates within 50 days after SIAs (Fig. 3a). Previous vaccine challenge studies found similar shedding durations in primary vaccine recipients, in whom shedding primarily occurred for 21–28 days, but sometimes extended for 49 days or more after dosage [5]. However,

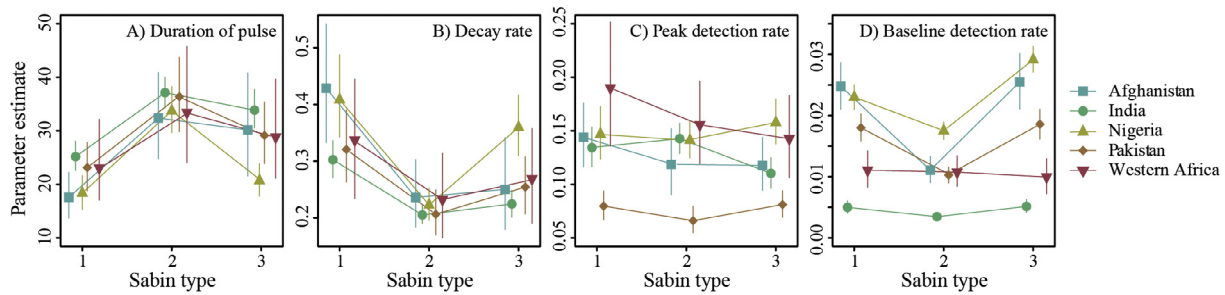


Fig. 3. Summary of parameter estimates for pre-cessation Sabin detection for the duration of the pulse in Sabin detection rates after SIAs (A), the decay in detection rates (B; an exponential-like rate of decay), peak detection rates (C), and baseline detection rates long times after SIAs (D). Panel A is in units of days. Panel B is in units of days⁻¹. Panel C and D are in units of Sabin positives * days⁻¹. Vertical lines represent 95% credible intervals.

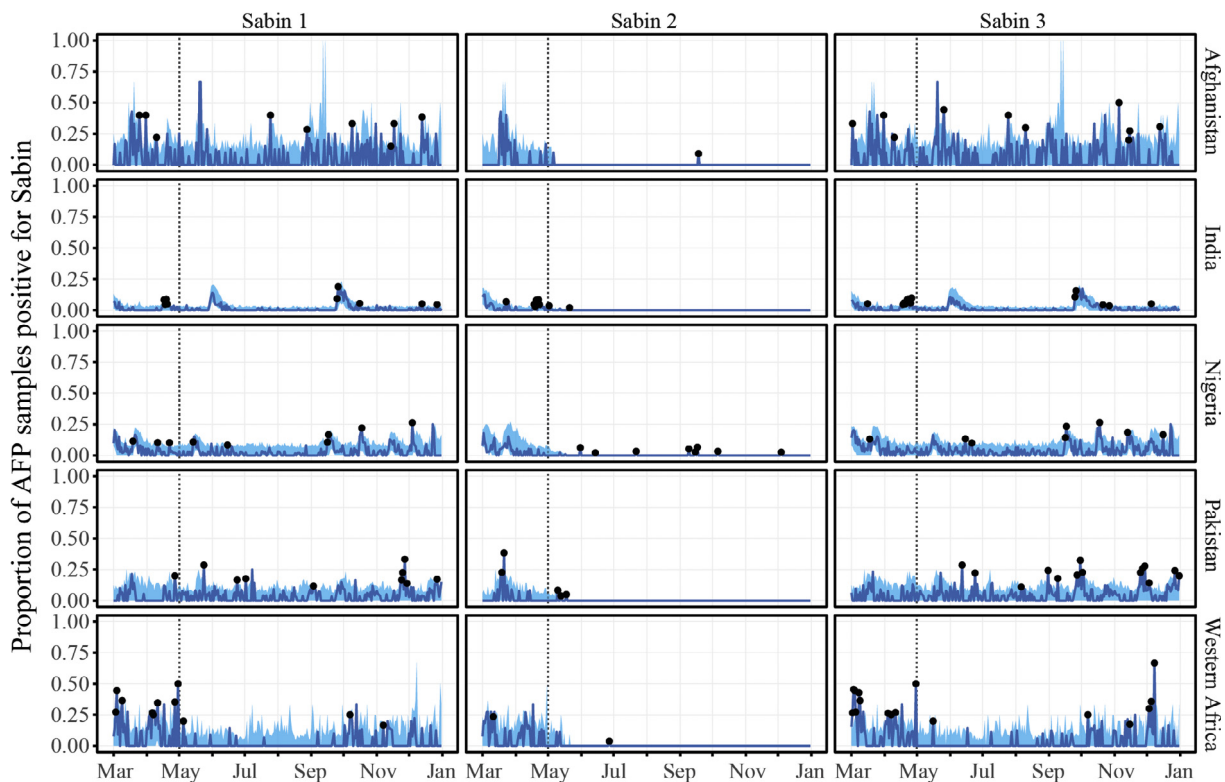


Fig. 4. Predicted and observed post-cessation Sabin detection rates. Solid blue lines indicate the daily proportion of Sabin (SL1-3) positive AFP samples. Light blue bands indicate 95% prediction intervals. Black dots indicate observed detection rates that fall outside of the 95% prediction intervals. The dotted vertical lines indicate the scheduled date of OPV2 cessation, April 30, 2016. For these predictions, we decayed baseline detection rates after OPV2 cessation to account for the removal of OPV2 from RI (section G in the Supplemental Material). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

our data also suggest that non-trivial amounts of SL detection result from secondary transmission since 3.6% of the SL positive cases in our study reported zero previous OPV doses (Fig. B in the Supplemental Material). While dose recall errors occur [22], these data are compatible with previous studies showing considerable secondary transmission of SL after SIAs and RI [15,23,24].

Our models demonstrated that SL2 had longer pulses in detection rates and slower rates of decay than SL1 and 3 (Fig. 3). This may be caused by lower type 2 immunity since OPV2 campaigns occurred at approximately half the frequency of OPV1 and 3 campaigns (Table 1) [18]. Low type 2 immunity would allow for increased secondary transmission of SL2 and prolonged shedding duration, since shedding duration is inversely related to prior vaccine exposure [1]. Regardless of immunity, previous studies have shown SL2 is more transmissible than SL1 and 3 [1]. For example, studies have found higher seroprevalence of type 2 than type 1 and

3 in communities without a recent history of vaccination or natural infection, suggesting that SL2 can spread widely throughout communities with low immunity [25,26].

4.2. Variation in detection rates between countries

We observed considerable variation in peak and baseline detection rates among countries (Fig. 3), but determining the cause is challenging. For example, peak detection rates in Pakistan were half those in Nigeria and western Africa. Low peak detection rates could result from low campaign coverage, low vaccine take, a poorly functioning AFP surveillance system, or high immunity. However, low campaign coverage in Pakistan seems unlikely since Pakistan had more reported doses than Nigeria and western Africa despite similar SIA rates (Table 1). Likewise, differences in baseline detection rates, like we observed in India and Nigeria, could exist

for disparate reasons. Low baseline detection rates could result from low RI coverage, or could conversely be caused by very high RI and SIA coverage, as high immunity would limit secondary transmission and shedding duration.

4.3. Initial assessment of OPV2 cessation

As predicted, SL2 detection rates have plummeted, but there have been 19 cases that deserve consideration (Fig. 4). The six detections in India and Pakistan are less concerning because they occurred within 50 days after cessation, and are compatible with shedding from primary or secondary contact with OPV2 just prior to cessation. The detection in Guinea is more concerning since Guinea had a large cVDPV2 outbreak in December 2015 (7 cVDPV2 cases) suggesting the conditions are present for prolonged type 2 circulation or unsanctioned tOPV use. Afghanistan also had a detection 141 days after cessation, likely from prolonged transmission or unsanctioned tOPV use. Subsequent field investigations have found evidence of unsanctioned tOPV use in a few instances in Afghanistan, India, Nigeria, and Pakistan [27].

More concerning is the situation in Nigeria, where multiple cVDPV2 lineages were detected in Borno just prior to OPV2 cessation after circulating undetected since 2014 [28]. The discovery of these relatively old and previously undetected cVDPV2s suggests gaps in AFP surveillance in one of the historic strongholds for polio. With surveillance gaps, cVDPV2 could circulate for years after cessation before detection. In response to the cVDPV2 detections, multiple emergency mOPV2 campaigns were conducted across northern Nigeria and parts of Cameroon, Chad, and Niger. These campaigns likely sourced some of the post-cessation SL2 detections in Nigeria. However, long-distance transmission of SL2 would have been required since 5 of the 11 SL2 detections in Nigeria (in Delta, Enugu, Katsina, and Kebbi states) occurred at least 200–450 km from the nearest mOPV2 campaigns. Alternatively, unsanctioned tOPV use or long-term local SL2 transmission (secondary transmission occurring >50 days after OPV2 use) could explain these detections. If long-distance or long-term transmission are to blame, this would not bode well for the risk of future cVDPV2 emergence: both are indicative of low immunity and the potential for cVDPV2 spread when type 2 immunity should be at its highest after cessation.

While our models ignore immunity, they were parameterized during an era of high type 2 immunity. Post-cessation, new births will cause type 2 immunity to decline in the 0–5 year-old cohort as much as 20% per year. As immunity falls, shedding duration and the potential for transmission will increase such that future OPV2 use, either used to stop outbreaks or unsanctioned, could seed new cVDPV2 outbreaks that ultimately require a global OPV2 reintroduction.

4.4. Molecular clock of OPV2

The continual genetic evolution of poliovirus implies that after 3–4 months of circulation, any remaining OPV2 strains are more than 50% likely to have acquired the 6 nucleotide substitutions to be classified as a VDPV2 instead of SL2 [29]. Thus, any remaining OPV2 strains are increasingly likely to be classified as VDPV2 instead of SL2, and future SL2 detections would imply unsanctioned tOPV use or long-range transmission from outbreak response initiatives.

4.5. Factors influencing prevalence estimates

We assumed the NP-AFP cases with Sabin-like virus included in our models were not VAPP cases. However, VAPP diagnosis is challenging and partially based on clinical symptoms. Misclassified

VAPP cases could inflate our prevalence estimates. However, VAPP is extremely rare (~1 case per 900,000 first doses of OPV) and thus unlikely to significantly influence our results [19].

As part of the vaccine switch from tOPV to bOPV, Afghanistan, India, Nigeria, Pakistan, and parts of western Africa began introducing one dose of IPV into RI in 2015–2016 in addition to the existing three OPV doses. While additional IPV can boost mucosal immunity and thus reduce acquisition and shedding duration of a subsequent poliovirus exposure, we did not compare detection rates before and after IPV introduction. The impacts of IPV are likely small on our results due to the limited time frame of introduction, the highly variable RI rates within and between countries, and the fact that RI only reaches 0–1 year olds which are uncommon in AFP surveillance (Fig. A in the Supplemental Material).

5. Conclusion

Our models establish benchmarks for the duration of SL detection after mass vaccination activities and serve as an assessment of OPV2 cessation. Our models also provide a framework for post-outbreak response monitoring for emergency mOPV2 use. For example, SL2 detections more than 50 days after a campaign could be used to flag regions that may have prolonged transmission due to low immunity and warrant close monitoring for potential cVDPV2 outbreaks, or alternatively indicate potential unsanctioned tOPV use that warrants field investigation. Environmental surveillance will also be vital for assessing OPV2 cessation since it often outperforms AFP surveillance for detection of poliovirus circulation [30].

Initial results for OPV2 cessation are mixed: SL2 detection rates have plummeted, except in Nigeria where mOPV2 use is ongoing in response to recent cVDPV2 detections. Moreover, the discovery of cVDPV2 lineages in Nigeria that circulated undetected for over 2 years highlights surveillance gaps. Since cVDPV2s have been known to circulate silently for years [31], it may be several years before we can declare the world free of circulating type 2 poliovirus.

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Contributors

All authors contributed to the conception and design of the study. SJK, MF, and HL prepared the statistical models and analyzed the data. KAM prepared the matched filter analysis. All authors contributed to the interpretation, writing, reviewing, and approval of the manuscript before submission.

Conflicts of interest

The authors have no conflicts of interest to declare.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.vaccine.2017.08.048>.

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